

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

November 2013

IT207 Computer Networking-1

Date: 13.11.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Define the term: "Data Communication". Which are the different applications of data communication network? [04]

(b) List and explain the fundamental characteristics of data communication. [03]

Q - 2 (a) Topology refers to the way a network is laid out. Explain the following topology with its advantages and disadvantages: [06]

- 1) Mesh Topology
- 2) Star Topology
- 3) Tree Topology

(b) Draw and explain the TCP/IP Protocol suite. [05]

OR

(b) List and Explain the different categories of computer network with example. [05]

(c) Explain the following term: [03]

- 1) Multipoint line configuration
- 2) Half-Duplex communication
- 3) Protocol

Q - 3 (a) Explain the functions of following layers of OSI model in detail: [08]

- 1) Data link Layer
- 2) Network Layer
- 3) Transport Layer
- 4) Session Layer

(b) What is flow control in data communication network? Which are the different methods of flow control? Explain each one in brief. [06]

OR

(b) How ENQ/ACK and Poll/Select methods of data link control works? Explain in detail. [06]

SECTION – II

- Q - 4 (a) What are the differences between Synchronous and Asynchronous transmission? [04]
(b) What are the advantages and disadvantages of optical fiber cable? [03]
- Q - 5 (a) Explain the terms: "Bit Interval" and "Bit Rate" of digital signals. [03]
(b) How Unipolar, Polar and Bipolar encoding techniques works? [06]

OR

- (b) Explain the working of amplitude shift keying (ASK) and Frequency shift keying (FSK) in communication network. [06]
(c) Explain the working of vertical redundancy check (VRC) technique with suitable example. [04]
- Q - 6 (a) Answer the following questions. (Attempt any three) [15]
1) Explain CRC Technique with suitable example.
2) Explain hamming code technique.
3) What is automatic repeat request (ARQ)? Explain ARQ techniques in brief.
4) Explain the nyquist theorem in detail.
5) Explain the Time-division multiplexing process in detail.
